TCC805x MCU BSP

API Specification for Local Interconnect Network

Rev. 1.01 [A]

2021-01-22

TABLE OF CONTENTS

Contents

Figures

Tables

1 INTRODUCTION

This document describes the local interconnect network device driver (hereinafter referred to as LIN driver) of TCC805x MCU BSP. The LIN is a serial communications protocol which efficiently supports the control of mechatronics nodes in distributed automotive applications. For more detailed information of the Local Interconnect Network, refer to ***"ISO standard ISO 17987 Part 1-7"***.

Telechips does not have LIN Bus hardware, but user can operate LIN communication using UART (Universal Asynchronous Receiver/Transmitter) interface hardware and external LIN transceiver. For more detailed information of the UART interface hardware in the TCC805x MCU Sub-System, refer to Chapter 12 Universal Asynchronous Receiver/Transmitter (UART) of the ***"PART 11-MICOM SUB-SYSTEM"*** in the ***"TCC805x Full Specification"***. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

LIN	Local Interconnect Network
PID	Protected Identifier
UART	Universal Asynchronous Receiver/Transmitter

3 SOURCE FILES

3.1 Location of Source Files

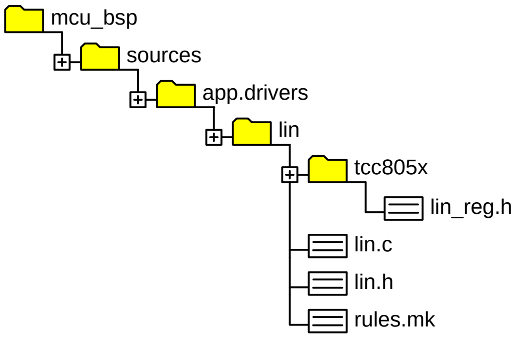

Figure 3.1 Location of Source Files

3.2 Simple Description of SOURCE FILES

- lin_reg.h
  - Define variables that are dependent on hardware
- lin.c
  - LIN driver source
- lin.h
  - LIN driver API header
- rules.mk
  - Rules used for build

4 CONSTANTS

4.1 Return Values

This Return values indicate function results.

Declaration

```
#define LIN_OK        (0)
#define LIN_ERROR     (-1)
```

Description

Value	*Description*
LIN_OK	*No error*
LIN_ERROR	*LIN error occurred*

Remark

None

4.2 LIN Data Size

Constant values for maximum size of LIN data

Declaration

```
#define LIN_MAX_DATA_SIZE     (8U)
```

Description

Value	*Description*
LIN_MAX_DATA_SIZE	*Maximum size of LIN data*

Remark

None

5 STRUCTURES

5.1 LINConfig_t

This struct includes configuration value for LIN driver.

Declaration

```
typedef struct LINConfig
{
    uint32     uiLinCh;
    uint32     uiPortSel;
    uint32     uiBaud;
    uint32     uiLinSlpPin;
} LINConfig_t;
```

Description

Value	Description
uiLinCh	UART Controller number (0 to 5)
uiPortSel	Channel to Port Mapping Information of UART (0 to 47). This value is saved in Port Selection Register.
uiBaud	Baud rate of LIN communication
uiLinSlpPin	GPIO Port used as LIN Transceiver Sleep Control Pin

Remark

None

6 APIs

6.1 LIN_Init

Initializes the LIN driver, sets the baud rate, UART channel, UART port selection, and wakes up the LIN transceiver.

Declaration

```
sint8 LIN_Init
(
    LINConfig_t    sLinCfg
);
```

Parameters

Value	Description
sLinCfg	[in] The specific configuration. Refer to the LINConfig_t.

Return

Value	Description
LIN_OK	Success
Others	Failure

Remark

None

6.2 LIN_SendData

Sends data to frame identifier.

Declaration

```
sint8 LIN_SendData
(
    LINConfig_t    sLinCfg,
    uint8          ucId,
    uint8 *        pucData,
    uint8          ucSize
);
```

Parameters

Value	Description
sLinCfg	[in] The specific configuration. Refer to the LINConfig_t.
ucId	[in] ucId is a frame identifier
pucData	[in] Data Array for sending. Array size is up to 8.
ucSize	[in] The number of sending data

Return

Value	Description
LIN_ERROR	Failure
LIN_OK	Success

Remark

None

6.3 LIN_ReadData

Reads data from slave that have specific frame identifier.

Declaration

```
sint8 LIN_ReadData
(
    LINConfig_t    sLinCfg,
    uint8          ucId,
    uint8 *        pucData
);
```

Parameters

Value	Description
sLinCfg	[in] The specific configuration. Refer to the LINConfig_t.
ucId	[in] ucId is a frame identifier
pucData	[out] Data Array for Read. Array size is up to 8.

Return

Value	Description
LIN_ERROR	Failure
Others	The number of read data

Remark

None

6.4 LIN_TransceiverWakeup

Sets LIN Transceiver status.

Declaration

```
sint8 LIN_TransceiverWakeup
(
    LINConfig_t    sLinCfg,
    boolean        bStatus
);
```

Parameters

Value	Description
sLinCfg	[in] The specific configuration. Refer to the LINConfig_t.
bStatus	[in] Set LIN Transceiver operation status. SALEnabled: Set Lin Transceiver as Wake up SALDisabled: Set Lin Transceiver as Sleep mode

Return

Value	Description
LIN_OK	Success
Others	Failure

Remark

None

7 How to use LIN Driver

7.1 Initialize LIN Driver

```
{
    ...

    LINConfig_t sLinConfig;

    sLinConfig.uiLinCh         = UART_CH2;
    sLinConfig.uiBaud          = 9600U;
    sLinConfig.uiPortSel       = 36U;
    sLinConfig.uiLinSlpPin     = GPIO_GPK(8UL);

    LIN_Init(sLinConfig);

    ...
}
```

7.2 Write Data

```
{
    ...
    uint8 ucSendData[3] = {0xAA, 0xBB, 0xCC};
    uint8 uiSendId     = 0x11;
    uint8 ucSendSize   = 3;

    uiStatus = LIN_SendData(sLinConfig, uiSendId, ucSendData, ucSendSize);

    LIN_SendData(sLinConfig, uiSendId, ucSendData, ucSendSize);

    ...
}
```

7.3 Read Data

```
{
    ...

    uint8 ucReadData[LIN_MAX_DATA_SIZE] = {0,};
    uint8 uiRendId        = 0x08;
    sint8 cReadSize;

    cReadSize = LIN_ReadData(sLinConfig, uiRendId, ucReadData);

    ...
}
```

8 REFERENCES

[1] TCC805x Full Specification

[2] ISO standard ISO 17987 Part 1-7

[3] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.01: 2021-01-22

- Updated
  - Chapter 5.1: Declaration
  - Chapter 7.2 and Chapter 7.3: Source Code

- Added
  - Chapter 4.2: LIN Data Size

Rev. 1.00: 2020-11-12

- First version release